

Vichaksha Viduranga Geekiyanage

ASSOCIATE AI/ML ENGINEER · PRODUCTION CONVERSATIONAL AI · COLOMBO, SRI LANKA

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Profile

Associate AI/ML Engineer with a year of industry experience building **production grade conversational AI**, promoted from an internship within the first year. Ships LLM agents on the **Meta WhatsApp Cloud API** in Sinhala, Tamil and English, with structural **guardrails** against hallucination, **hybrid RAG** retrieval and serverless deployment on **Google Cloud Run**. Published in **Scopus indexed** proceedings. Fluent English, native Sinhala.

Technical Skills

LLM & Agents	Prompt Engineering, LLM Guardrails, Hallucination Prevention, RAG, Hybrid RAG, Function/Tool Calling, Agent Design, LLM Evaluation, Google Gemini, Gemini Live API, Vertex AI, Azure AI Foundry, Azure AI Speech, OpenAI API, LangChain, LLM Performance Monitoring (LangSmith, Fiddler AI)
Machine Learning	PyTorch, TensorFlow, Hugging Face Transformers, PEFT/LoRA Fine-Tuning, Whisper ASR, Sentence Transformers, YOLO, OpenCV
Backend & APIs	Python, FastAPI, Flask, Node.js, Java, REST APIs, WebSockets, Webhooks, Meta WhatsApp Cloud API
Data & Retrieval	PostgreSQL, MongoDB, Redis, Qdrant (Vector Database), Elasticsearch, FAISS, Hybrid Search (BM25 + Dense Embeddings), Re-ranking
Cloud & DevOps	Google Cloud Run, Google Cloud Platform (GCP), Serverless, Terraform (IaC), Docker, GitHub Actions (CI/CD), Cloud Logging & Monitoring, Linux, Git
Security	Input Validation, SQL Injection Prevention, JWT Authentication, API Key & Secret Management, Rate Limiting, Circuit Breakers, Idempotency, Automated Testing

Professional Experience

Surge Robotics (formerly Idea8 Pvt Ltd), **Colombo**

Sep 2025 – Present

Associate AI/ML Engineer · promoted from AI/ML Intern

Production Conversational AI Agent - Meta WhatsApp Cloud API

- Built and **operate in production today**: an LLM agent handling ordering, pricing, order status and human escalation across 10K conversations a day for live customers in Sinhala, Tamil and English.
- Designed a structural **guardrail layer** blocking any outbound message containing a fact not sourced from code or a verified store, eliminating hallucinated prices, phone numbers and order IDs.
- Grounded answers in **hybrid RAG**: BM25 in **Elasticsearch** fused with dense **sentence transformer** embeddings served from self-hosted **Qdrant** (HNSW index, payload filtering, snapshots), re-ranked passages being the only admissible source of facts.
- **Securely deployed** as a containerised service: authenticated webhooks, secrets outside version control, least privilege service accounts, CI/CD with tagged rollbacks.
- **Multi-provider LLM failover** with per provider circuit breakers and recovery probing, over **Redis** state: history, carts, deduplication, takeover locks, TTL expiry.
- **Observability and LLM performance monitoring** in **Cloud Logging**, **Cloud Monitoring**, **LangSmith** and **Fiddler AI**: structured logs with correlation IDs, token and latency metrics, alerting on error rate and p95; cut LLM cost 30% with prompt caching and a deterministic cache.
- Authored a **603 test regression suite**; led incident response and the postmortem for a production outage.

Garment Measurement Computer Vision - current project (Surge Robotics)

- Leading an automated quality control system targeting **1 mm accuracy** from a single camera: **YOLO pose estimation** across 14 keypoints, **BiRefNet** segmentation and **Charuco** calibration, on a dataset collected and annotated in-house.

Multilingual Voice AI Assistant

- Deployed a real-time voice agent on **Google Cloud Run** (FastAPI + Gemini Live API): bidirectional audio streaming, barge-in handling, tool calling into booking back ends, and **RAG** retrieval over a bilingual knowledge base with an async rate limiter on model quota.
- Worked across both cloud stacks: **Azure AI Foundry** for model hosting and agent deployment, with **Azure AI Speech** for speech to text and text to speech.

Publications

Enhancing Copyright Detection through Lyrics Analysis in Sinhala Song Audio Sep 2026
5th SLIIT International Conference on Engineering and Technology (SICET 2026), Malabe, Sri Lanka
Published in **Scopus indexed** conference proceedings as first author.

Selected Projects

Dialix - Voice and WhatsApp AI Agent Platform (own product) 2026 – Present
Gemini Live API, FastAPI, Qdrant, Terraform, Google Cloud Run Visit dialix.dev

- Commercial voice and messaging agent for Sri Lankan businesses: answers in Sinhala and English, handles ordering and booking flows against the business's own catalogue, and escalates to a human on request.
- **Serverless architecture** on **Google Cloud Run**: containerised FastAPI services, request based autoscaling and scale to zero between calls, so there is no instance to keep warm or patch.
- **Infrastructure as code with Terraform**: services, IAM, secrets and registries in version control, so a new environment is a reviewed plan and apply.

Sinhala Lyrical ASR (NLP) Aug 2024 – Present
Whisper, LoRA, Elasticsearch · 

- Fine-tuned Whisper-large-v2 with LoRA for Sinhala lyric transcription; captured harmonies in heavy metal music, especially in Sinhala music, with low VRAM usage.
- Trained to convergence over **20,244 steps**, loss 2.17 → 0.12; Elasticsearch lyric search over vector embeddings underpins the SICET 2026 publication above.

Education

BSc (Hons) Electronic & Computer Science 2026
University of Kelaniya, Sri Lanka Second Class

G.C.E. Advanced Level - Physical Science Stream Nalanda College, Colombo 10

Certifications

LinkedIn Learning, 2026: Building AI Agents: From Zero to Production (Microsoft authored) · RAG and Fine-Tuning Explained · Google Cloud Build: Test, Deploy and Manage Code
Other: Prompt Engineering (IBM Skills Network, 2025) · Makerspace Training (American Center Colombo, 2025)

Leadership & Volunteering

- **Secretary**, IEEE IES Student Branch Chapter, UoK (2024) — events and workshops for 100+ peers.
- **Speaker**, IEEE IES AI Challenge (2026) — talk on building AI agents for the student teams taking part.
- **Coordinator**, IEEE Robot Battle UoK (2024) — robotics competition, 200+ participants, IEEE sponsorship secured.

References

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